Examiner only

2. (a) A class investigates radioactive decay.
They model decay using **8-sided** dice.

Each of the 10 groups has 50 dice.
They throw the dice and remove all which land with an 8 facing upwards.
These represent decayed nuclei.
They repeat 7 more times and record the number of dice remaining after each throw.
Each group's results are then added together.

(i) Freya suggests that it is good practice to add the results together to give a larger sample size.
Explain whether you agree. [1]

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(ii) The teacher calculates that after the first throw **around** 440 of the 8-sided dice should remain out of the 500.
Explain how she determined this number. [2]

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(b) The results from the experiment are given in the table below.

Number of throws	Number of dice remaining
0	500
1	435
2	380
3	335
4	291
5	256
6	224
7	196
8	171

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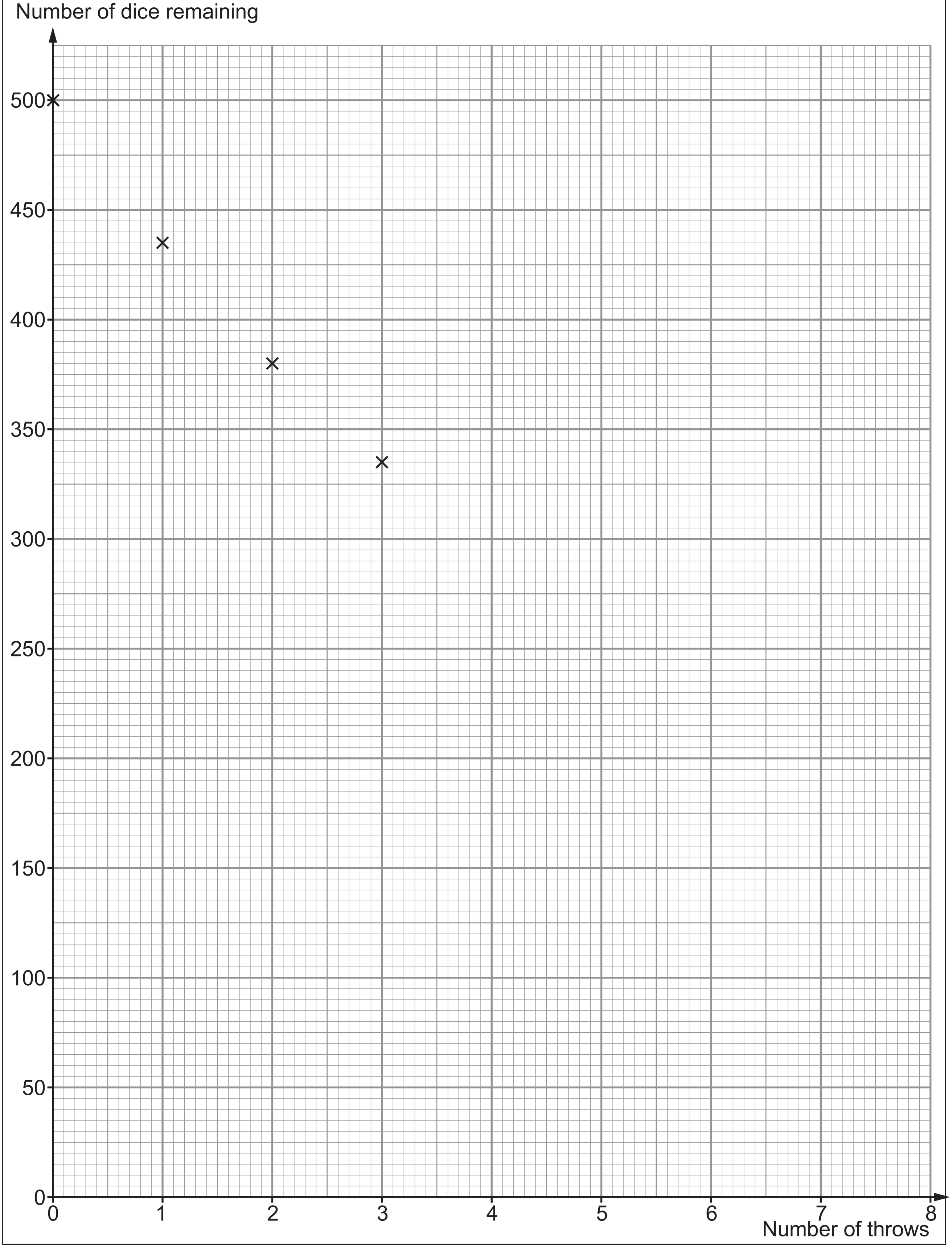
(i) Plot the data on the grid below and draw a suitable curve.
The first 4 points have been plotted for you. [3]



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(ii) **Add lines to the graph** to find the number of throws required to halve the number of dice. This is the half-life.
Give your answer to 1 decimal place. [2]

number of throws = ...

(iii) The experiment is repeated with 10-sided dice.
Ieuan suggests the data could be used to model nuclear decay with a shorter half-life than with 8-sided dice.
Explain whether you agree. [2]

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